

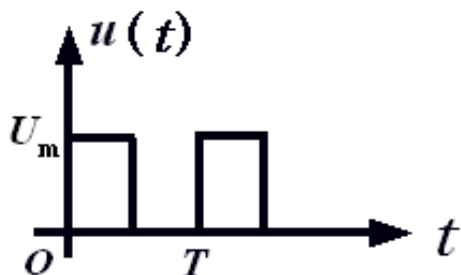
第十三章 非正弦周期信号电路

- 周期信号的傅里叶级数
- 非正弦周期信号的平均值、有效值，平均功率
- 非正弦周期信号电路的计算

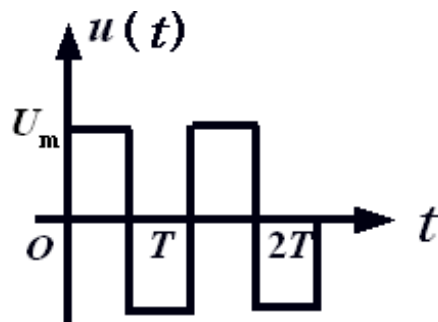
§ 13-1 非正弦周期信号

概述

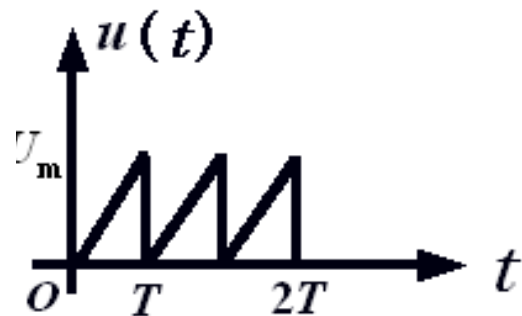
一、各种非正弦周期信号



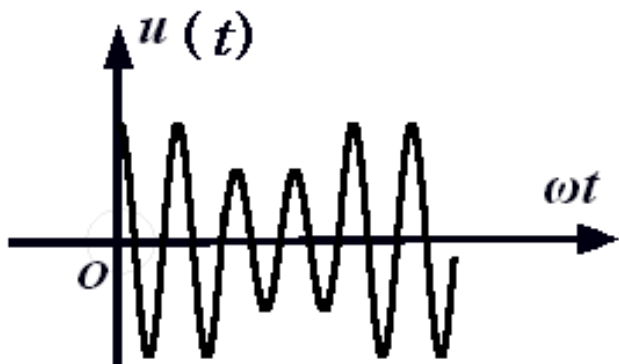
单向周期性脉冲信号



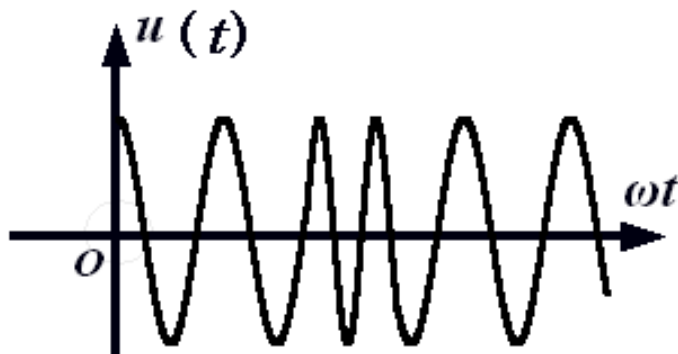
双向周期性脉冲信号



锯齿扫描信号



调幅信号

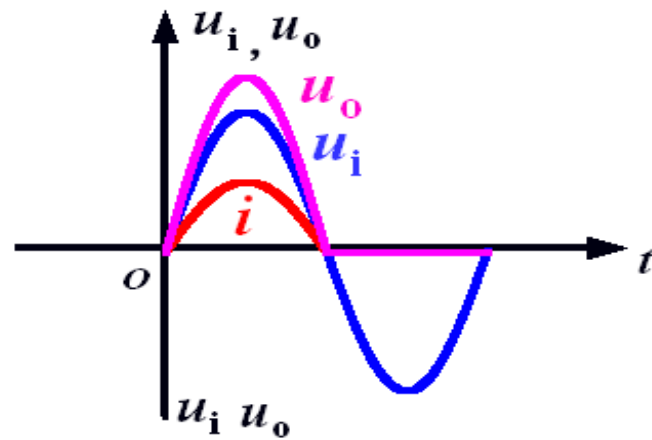
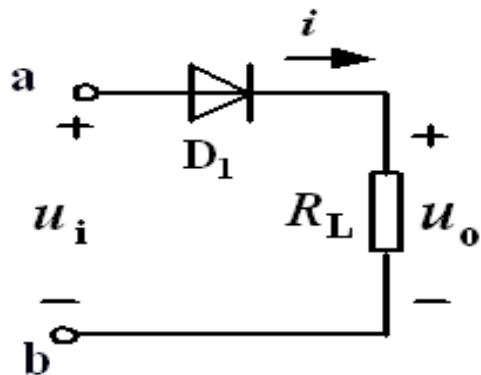


调频信号

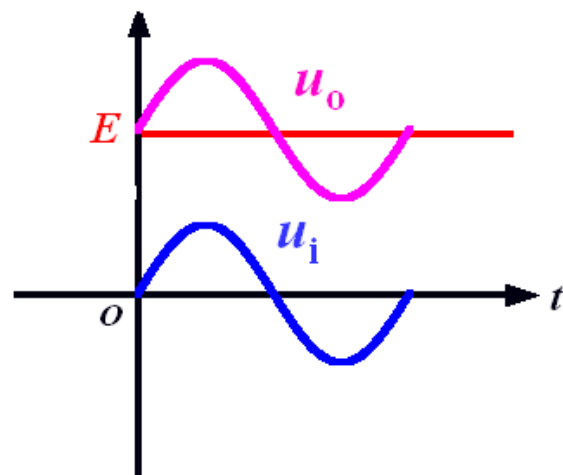
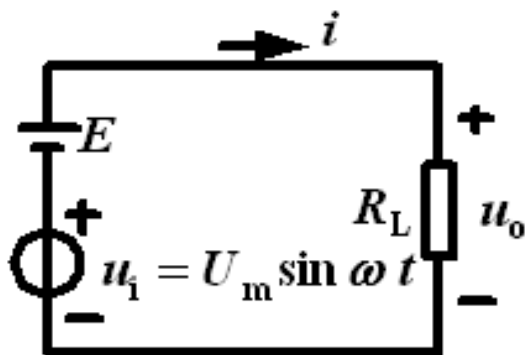
§ 13-1 非正弦周期信号

非正弦周期信号产生的原因

1. 电路中含有非线性元件



2. 电路中含有不同频率的激励源



§ 13-2 非正弦周期信号的傅里叶级数

$$f(t) = f(t + nT) = \frac{a_0}{2} + \sum_{k=1}^{\infty} [a_k \cos k\omega_1 t + b_k \sin k\omega_1 t]$$

$$A_{km} = \sqrt{a_k^2 + b_k^2} \quad \varphi_k = \arctan\left(\frac{-b_k}{a_k}\right) \quad \omega_1 = \frac{2\pi}{T}$$

$$a_k = A_{km} \cos \varphi_k \quad b_k = -A_{km} \sin \varphi_k \quad A_{km} e^{j\phi_k} = a_k - jb_k$$

$$\sum_{k=1}^{\infty} [a_k \cos k\omega_1 t + b_k \sin k\omega_1 t]$$

$$= \sum_{k=1}^{\infty} A_{km} [\cos \varphi_k \cos k\omega_1 t - \sin \varphi_k \sin k\omega_1 t]$$

$$= \sum_{k=1}^{\infty} [A_{km} \cos(k\omega_1 t + \varphi_k)]$$

$$f(t) = \frac{A_0}{2} + \sum_{k=1}^{\infty} A_{km} \cos(k\omega_1 t + \phi_k)$$

§ 13-2 非正弦周期信号的傅里叶级数

$$\frac{1}{T} \int_0^T f(t) \cos(n\omega_1 t) dt =$$

$$\frac{1}{T} \int_0^T \left\{ \frac{a_0}{2} + \sum_{k=1}^{\infty} [a_k \cos k\omega_1 t + b_k \sin k\omega_1 t] \right\} \cos(n\omega_1 t) dt$$

$$\frac{1}{T} \int_0^T f(t) \sin(n\omega_1 t) dt =$$

$$\frac{1}{T} \int_0^T \left\{ \frac{a_0}{2} + \sum_{k=1}^{\infty} [a_k \cos k\omega_1 t + b_k \sin k\omega_1 t] \right\} \sin(n\omega_1 t) dt$$

§ 13-2 非正弦周期信号的傅里叶级数

$$f(t) = f(t + nT) = \frac{a_0}{2} + \sum_{k=1}^{\infty} [a_k \cos k\omega_1 t + b_k \sin k\omega_1 t]$$

$$a_k = \frac{2}{T} \int_0^T f(t) \cos(k\omega_1 t) dt = \frac{1}{\pi} \int_0^{2\pi} f(t) \cos(k\omega_1 t) d(\omega_1 t)$$

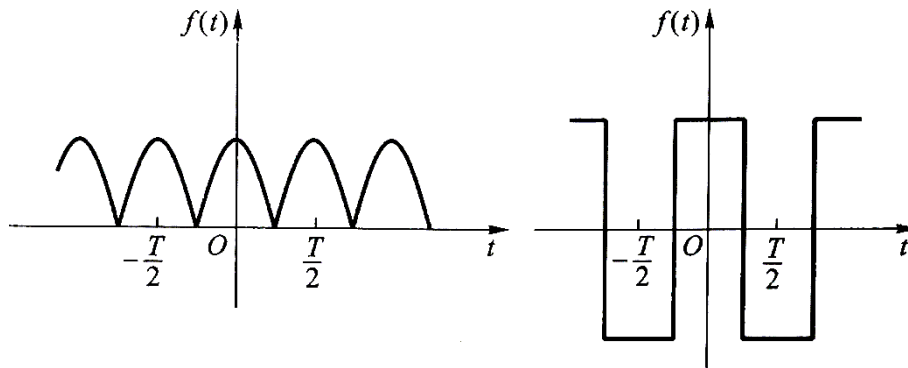
$$b_k = \frac{2}{T} \int_0^T f(t) \sin(k\omega_1 t) dt = \frac{1}{\pi} \int_0^{2\pi} f(t) \sin(k\omega_1 t) d(\omega_1 t)$$

§ 13-2 非正弦周期信号的傅里叶级数

偶函数

$$f(t) = f(-t)$$

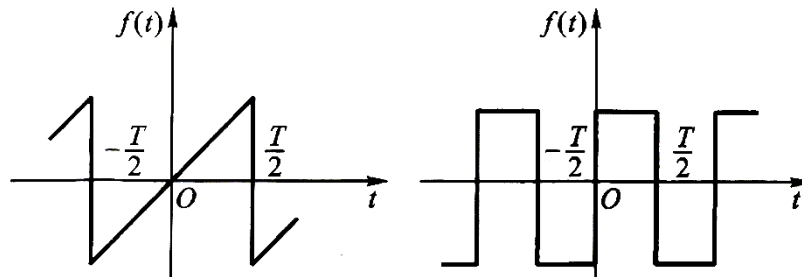
$$f(t) = \sum_{k=1}^{\infty} [a_k \cos k\omega_1 t]$$



奇函数

$$f(t) = -f(-t)$$

$$f(t) = \sum_{k=1}^{\infty} [b_k \sin k\omega_1 t]$$



§ 13-2 非正弦周期信号的傅里叶级数

$$f(t) = f(t + nT) = \frac{a_0}{2} + \sum_{k=1}^{\infty} [a_k \cos k\omega_1 t + b_k \sin k\omega_1 t] = \frac{A_0}{2} + \sum_{k=1}^{\infty} A_{km} \cos(k\omega_1 t + \phi_k)$$

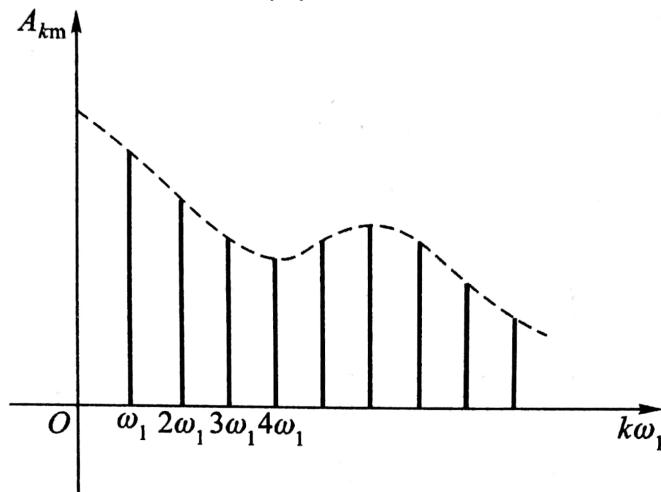
$$a_k = \frac{2}{T} \int_0^T f(t) \cos(k\omega_1 t) dt = \frac{1}{\pi} \int_0^{2\pi} f(t) \cos(k\omega_1 t) d(\omega_1 t)$$

$$b_k = \frac{2}{T} \int_0^T f(t) \sin(k\omega_1 t) dt = \frac{1}{\pi} \int_0^{2\pi} f(t) \sin(k\omega_1 t) d(\omega_1 t)$$

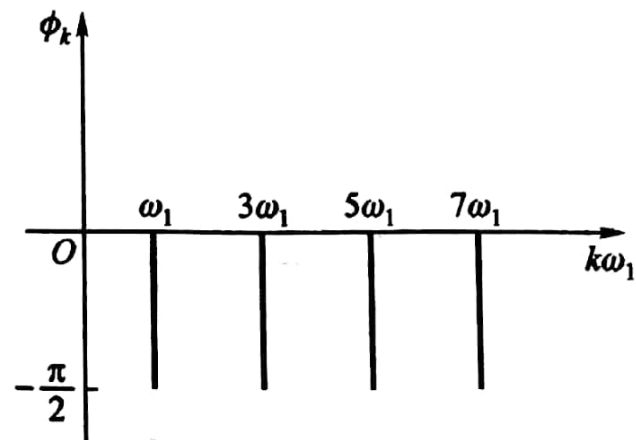
$$A_{km} = \sqrt{a_k^2 + b_k^2}$$

$$\varphi_k = \arctan \left(\frac{-b_k}{a_k} \right)$$

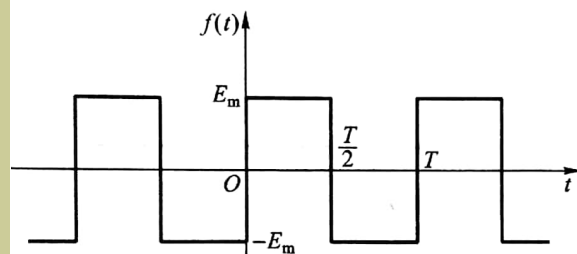
幅度频谱： $A_{km} - k\omega_1$



相位频谱： $\varphi_k - k\omega_1$



例13-1 矩形信号的傅里叶级数展开式及其频谱



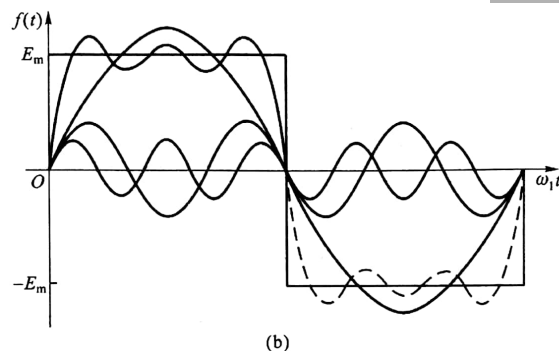
$$\begin{cases} f(t) = E_m & 0 \leq t \leq \frac{T}{2} \\ f(t) = -E_m & \frac{T}{2} \leq t \leq T \end{cases}$$

$$\begin{aligned} A_{km} e^{j\phi_k} &= a_k - jb_k = \frac{2}{T} \int_0^T f(t) e^{-jk\omega_1 t} dt \\ &= \frac{E_m}{jk\pi} (1 - e^{-jk\pi})^2 \end{aligned}$$

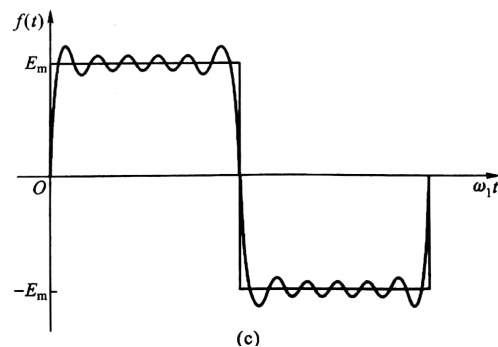
$$A_{km} = \frac{4E_m}{k\pi} \quad \phi_k = -90^\circ \quad k \text{ 为奇数}$$

得到傅里叶级数展开式:

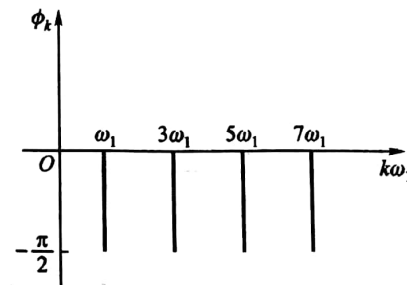
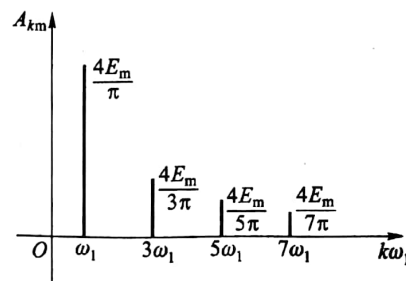
$$f(t) = \frac{4E_m}{\pi} \left[\sin(\omega_1 t) + \frac{1}{3} \sin(3\omega_1 t) + \frac{1}{5} \sin(5\omega_1 t) + \cdots \right]$$



5次谐波



11次谐波



§13.3 平均值、有效值、平均功率

一、有效值

$$I = \sqrt{\frac{1}{T} \int_0^T i^2 dt}$$

$$I = \sqrt{\frac{1}{T} \int_0^T [I_0 + \sum_{k=1}^{\infty} I_{km} \cos(k\omega_1 t + \varphi_k)]^2 dt}$$

$$I = \sqrt{I_0^2 + I_1^2 + I_2^2 + \cdots} = \sqrt{I_0^2 + \sum_{k=1}^{\infty} I_k^2}$$

$$U = \sqrt{U_0^2 + U_1^2 + U_2^2 + \cdots} = \sqrt{U_0^2 + \sum_{k=1}^{\infty} U_k^2}$$

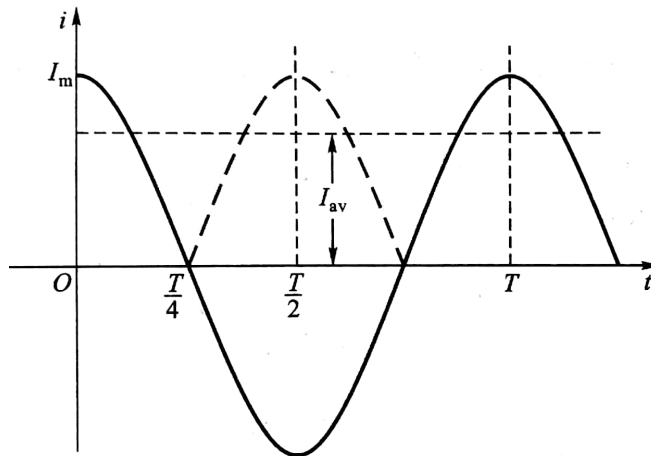
§ 13.3 平均值、有效值、平均功率

二、平均值 $f(t) = a_0 + \sum_{n=1}^{\infty} [A_n \cos(n\omega_1 t + \varphi_n)]$

$$I_{\text{av}} = \frac{1}{T} \int_0^T |i(t)| dt$$

$$I_{\text{av}} = \frac{1}{T} \int_0^T |I_m \cos(\omega t)| dt = \frac{4I_m}{T} \int_0^{T/4} \cos(\omega t) dt$$

$$= \frac{4I_m}{\omega T} \sin(\omega t) \Big|_0^{T/4} = \frac{2I_m}{\pi} = 0.637 I_m = 0.898 I$$



§ 13.3 平均值、有效值、平均功率

三、平均功率

$$u(t) = U_0 + \sum_{k=1}^{\infty} [U_{km} \sin(k\omega_1 t + \varphi_{ku})]$$

$$i(t) = I_0 + \sum_{k=1}^{\infty} [I_{km} \sin(k\omega_1 t + \varphi_{ki})]$$

$$P = \frac{1}{T} \int_0^T p dt = \frac{1}{T} \int_0^T ui \, dt$$

不同频率的正弦电压与电流乘积的积分为0

$$P = P_0 + \sum_{k=1}^{\infty} P_k = U_0 I_0 + \sum_{k=1}^{\infty} U_k I_k \cos(\varphi_{ku} - \varphi_{ki}) \quad U_k = \frac{U_{km}}{\sqrt{2}}, I_k = \frac{I_{km}}{\sqrt{2}}$$

四、视在功率

$$S = UI = \sqrt{U_0^2 + \sum_{k=1}^{\infty} U_k^2} \sqrt{I_0^2 + \sum_{k=1}^{\infty} I_k^2}$$

§ 13.3 平均值、有效值、平均功率

■ 注意：

- 1、不同频率的电压和电流不构成平均功率。
- 2、对于RLC串联回路, 在非正弦周期电压源的作用下, 直流分量不构成平均功率。
- 3、对于非正弦周期信号电路：

$$\begin{aligned} I &\neq I_0 + I_1 + I_2 + \dots & I &= \sqrt{I_0^2 + I_1^2 + I_2^2 + \dots} = \sqrt{I_0^2 + \sum_{k=1}^{\infty} I_k^2} \\ U &\neq U_0 + U_1 + U_2 + \dots & U &= \sqrt{U_0^2 + U_1^2 + U_2^2 + \dots} = \sqrt{U_0^2 + \sum_{k=1}^{\infty} U_k^2} \\ S &\neq U_0 I_0 + U_1 I_1 + U_2 I_2 + \dots & S &= UI = \sqrt{U_0^2 + \sum_{k=1}^{\infty} U_k^2} \sqrt{I_0^2 + \sum_{k=1}^{\infty} I_k^2} \end{aligned}$$

§13.4 非正弦周期信号电路的谐波分析法

方法：

- 1.将非正弦周期信号分解为直流分量和各次谐波分量之和。**
- 2.分别求直流分量和各次谐波分量单独作用于电路时所产生的分响应。**
- 3.将各响应的瞬时值进行叠加。**

注意：

各次谐波的感抗、容抗是相应角频率的函数。

$$X_L = k\omega_1 L \qquad X_C = \frac{1}{k\omega_1 C}$$

例13-2 求RLC串联电路中电流和电阻吸收的平均功率

$$R = 3\Omega, \frac{1}{\omega_1 C} = 21\Omega, \omega_1 L = 0.429\Omega,$$

$$u_s = [280.11\cos(\omega_1 t) + 93.37\cos(3\omega_1 t) + 56.02\cos(5\omega_1 t) + 40.03\cos(7\omega_1 t) + 31.12\cos(9\omega_1 t) + \dots] \text{ V}$$

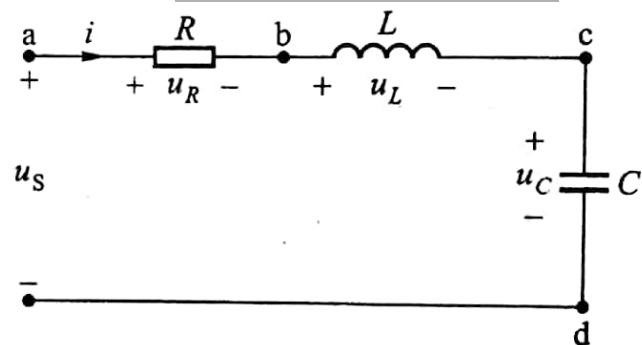
$$\textcircled{1} \quad \dot{I}_{m(k)} = \frac{\dot{U}_{sm(k)}}{Z(k\omega_1)} = \frac{\dot{U}_{sm(k)}}{R + jk\omega_1 L - j\frac{1}{k\omega_1 C}}$$

$$Z(k\omega_1) = 3 + j\left(0.429k - \frac{21}{k}\right) = 3\left[1 + j\left(0.143k - \frac{7}{k}\right)\right]$$

$$\begin{cases} \varphi_{(k)} = \arctan\left(0.143k - \frac{7}{k}\right) \quad (\text{阻抗角}) \\ \dot{I}_{m(k)} = \frac{1}{3}\cos\varphi_{(k)} \dot{U}_{sm(k)} \angle -\varphi_{(k)} \\ P_{(k)} = \frac{1}{2} I_{m(k)}^2 \cdot R = 1.5 I_{m(k)}^2 \end{cases}$$

$$\textcircled{2} \quad \begin{aligned} k=1 \quad & \varphi_{(1)} = -81.70^\circ (\text{容性}) \\ & \dot{I}_{m(1)} = 13.47 \angle 81.70^\circ \text{ A} \\ & P_{(1)} = 272.33 \text{ W} \end{aligned}$$

$$\begin{aligned} k=3 \quad & \varphi_{(3)} = -62.30^\circ (\text{容性}) \\ & \dot{I}_{m(3)} = 14.47 \angle 62.30^\circ \text{ A} \\ & P_{(3)} = 314.06 \text{ W} \end{aligned}$$



$$\begin{aligned} k=5 \quad & \varphi_{(5)} = -34.41^\circ (\text{容性}) \\ & \dot{I}_{m(5)} = 15.41 \angle 34.41^\circ \text{ A} \end{aligned}$$

$$P_{(5)} = 356.00 \text{ W}$$

$$\begin{aligned} k=7 \quad & \varphi_{(7)} = 0^\circ (\text{谐振}) \\ & \dot{I}_{m(7)} = 13.34 \angle 0^\circ \text{ A} \end{aligned}$$

$$P_{(7)} = 267.07 \text{ W}$$

$$\begin{aligned} k=9 \quad & \varphi_{(9)} = 26.99^\circ (\text{感性}) \\ & \dot{I}_{m(9)} = 9.24 \angle -26.99^\circ \text{ A} \end{aligned}$$

$$P_{(9)} = 128.17 \text{ W}$$

③

$$\begin{aligned} i &= 13.47\cos(\omega_1 t + 81.70^\circ) + 14.47\cos(3\omega_1 t + 62.30^\circ) + \\ & 15.41\cos(5\omega_1 t + 34.41^\circ) + \dots \end{aligned}$$

$$P = P_{(1)} + P_{(3)} + P_{(5)} + \dots + P_{(9)} = 1337.63 \text{ W}$$